

# Zion v0.11.0 (16c3a1b)

Performance and HTTP conformance report · measured on a dedicated host with a fixed procedure

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| Version        | zion 0.11.0 (unknown unknown)                                                  |
| Label          | v0.11.0 (16c3a1b)                                                              |
| Binary SHA-256 | 8e41bdfa918f73eb7502f74698f588cb...                                            |
| Date (UTC)     | 2026-10-08T20:31:25Z                                                           |
| Host           | Intel(R) Core(TM) i7-6700 CPU @ 3.40GHz, 4 CPUs, 8192 MiB, kernel 6.17.4-2-pve |
| Governor       | performance                                                                    |
| Core layout    | origin 0 · server 1 · load 2,3                                                 |
| Procedure      | 1 (harness 345e9242c65e)                                                       |
| Example site   | SHA-256 7873d1c9b2e29573518971eb...                                            |
| Trials         | 5 × 10 s (+ 3 s warm-up)                                                       |
| TLS            | nginx: TLSv1.3 TLS_AES_256_GCM_SHA384; zion: TLSv1.3 TLS_AES_256_GCM_SHA384    |

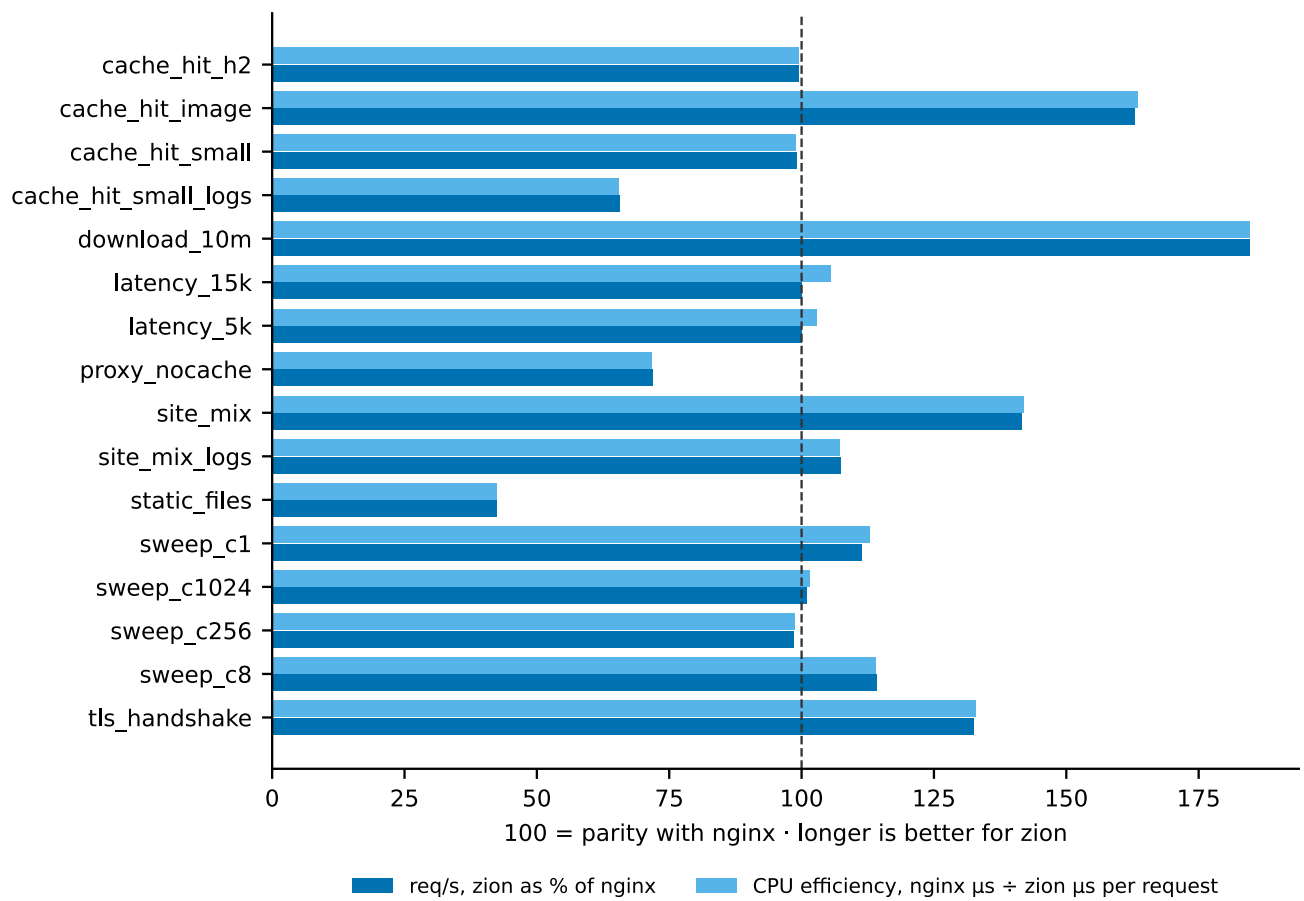
## Summary

The host moved during the run (the canary drifted 4.7% against a 3% tolerance): **treat these numbers as indicative and repeat the run.**

16 scenarios, 0 discarded trial(s) — sweep\_c1/nginx, sweep\_c1/zion did not keep the server at 80% CPU, so the figure is not a ceiling ( $\Delta$  in the table; with few connections that is expected). Each proxy ran on exactly one core, so every figure is per core. zion used less CPU per request than nginx in **10 of 16** scenarios.

| Headline scenario | zion req/s | nginx req/s | zion ÷ nginx | zion $\mu$ s/req | nginx $\mu$ s/req | zion p99 | nginx p99 |
|-------------------|------------|-------------|--------------|------------------|-------------------|----------|-----------|
| cache_hit_small   | 32.7k      | 33.0k       | 99%          | 30.6             | 30.2              | 2.30 ms  | 2.29 ms   |
| cache_hit_h2      | 41.7k      | 42.0k       | 99%          | 23.9             | 23.8              | -        | -         |
| site_mix          | 15.9k      | 11.2k       | 142%         | 62.8             | 89.1              | 6.30 ms  | 9.68 ms   |
| proxy_nocache     | 17.6k      | 24.6k       | 72%          | 56.4             | 40.5              | 5.03 ms  | 3.89 ms   |
| static_files      | 8.8k       | 20.7k       | 42%          | 113.9            | 48.4              | 17.00 ms | 3.52 ms   |
| tls_handshake     | 3.2k       | 2.4k        | 133%         | 308.8            | 410.3             | 11.20 ms | 1.75 ms   |

Access log off on both unless the scenario says `_logs`. “ $\mu$ s/req” is CPU time of the whole proxy process tree per request, from /proc.



## What was measured and how

An example web site (230 files, 25 MiB: a landing page and 40 documents, a JS/CSS bundle, fonts, 60 photographs, 120 JSON documents and a 10 MiB download) is generated from a fixed seed and served by a real nginx origin. **zion** sits in front of it; a real nginx with `proxy_cache` sits in front of the same origin as the reference; the origin read directly is the floor. All traffic is TLS on loopback, load generated by **wrk** (HTTP/1.1), **h2load** (HTTP/2) and **oha** (fixed arrival rate, latencies corrected for coordinated omission).

**Why the numbers can be trusted, and what “deterministic” means here.** The bytes served, the sequence of requests, the tool versions, the core layout and the order of every step are fixed and recorded above. The numbers themselves cannot be bit-identical on a real CPU; what is fixed is the size of the noise. Each scenario is repeated in 5 trials interleaved across the targets, so a slow drift of the host falls on all of them alike; the table gives the median and the minimum-maximum. A fixed CPU workload (the canary) brackets the run and a drift over 3% marks the whole report as noisy. Every trial is validated: any non-2xx response or transport error discards the trial, and a “cache hit” scenario is only reported if the origin shows it received (almost) no requests. The host must be at least 97% idle for five seconds before anything is timed.

**Limits, said plainly.** Loopback on one machine measures what the proxy costs per request, not a network. One core per proxy is a deliberate normalisation: it makes req/s a per-core figure and nginx’s `worker_processes 1` a fair partner, but with the access log on, **zion**’s log writer shares that core, which overstates the log’s cost relative to a multi-core deployment ( $\mu\text{s}/\text{req}$  is the figure to read). The CPU is shared with other guests of the same hypervisor and turbo frequency varies. HTTP/3, the WAF, rate limiting, response compression and WebSocket are not measured here.

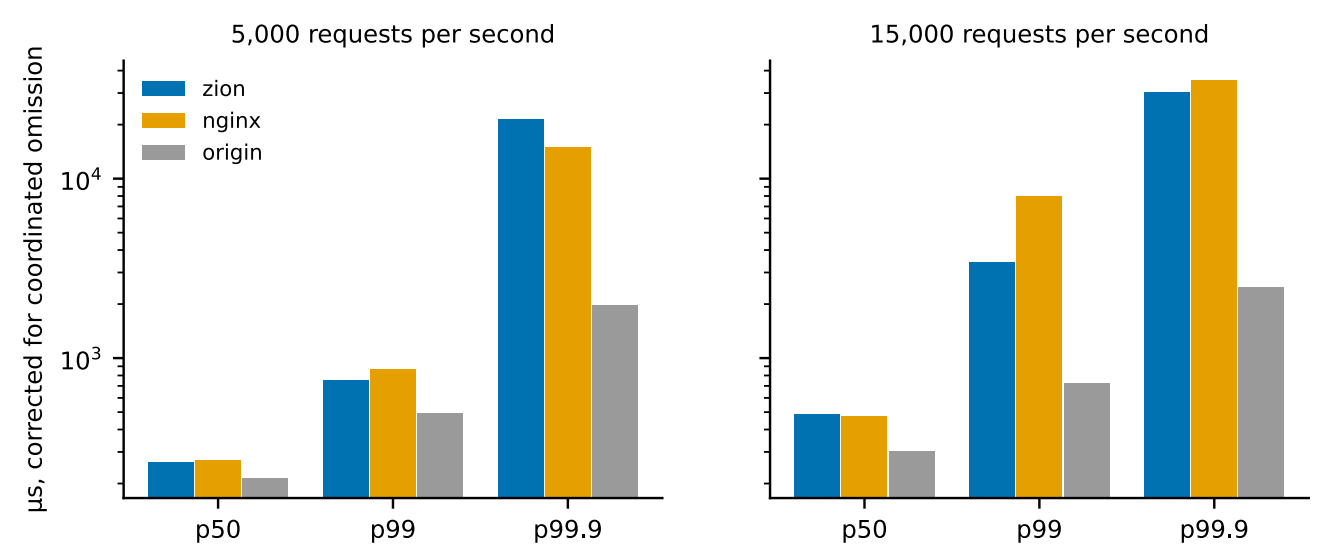
## Results by scenario

| target                                                                                                                                | req/s                                                                                     | min-max     | $\mu\text{s}$ CPU/req | p50               | p99               | p99.9     | server CPU | RSS MiB | cache hit |
|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------|-----------------------|-------------------|-------------------|-----------|------------|---------|-----------|
| <b>cache_hit_h2</b> · h2load · 16 conn · the same document over HTTP/2, 16 connections x 10 streams                                   |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 42.0k                                                                                     | 41.7k–42.6k | 23.8                  | –                 | –                 | –         | 100%       | 23      | 100.0%    |
| zion                                                                                                                                  | 41.7k                                                                                     | 41.2k–41.8k | 23.9                  | –                 | –                 | –         | 100%       | 15      | 100.0%    |
| <b>cache_hit_image</b> · wrk · 64 conn · GET a ~200 KB cached image: copy and TLS cost, not request cost                              |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 4.2k                                                                                      | 4.2k–4.2k   | 238.2                 | 15.17 ms          | 17.17 ms          | 19.59 ms  | 100%       | 25      | 100.0%    |
| origin                                                                                                                                | 23.0k                                                                                     | 22.9k–23.1k | 37.4                  | 1.42 ms           | 3.32 ms           | 10.84 ms  | 86%        | 12      | –         |
| zion                                                                                                                                  | 6.8k                                                                                      | 6.8k–6.9k   | 145.8                 | 9.43 ms           | 11.61 ms          | 16.04 ms  | 100%       | 15      | 100.0%    |
| <b>cache_hit_small</b> · wrk · 64 conn · GET a 5 KB cached JSON document, TLS, HTTP/1.1, 64 keep-alive connections                    |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 33.0k                                                                                     | 32.5k–34.6k | 30.2                  | 1.91 ms           | 2.29 ms           | 3.67 ms   | 100%       | 26      | 100.0%    |
| origin                                                                                                                                | 54.9k                                                                                     | 54.4k–57.3k | 18.2                  | 1.16 ms           | 1.43 ms           | 2.82 ms   | 100%       | 12      | –         |
| zion                                                                                                                                  | 32.7k                                                                                     | 32.3k–34.1k | 30.6                  | 1.97 ms           | 2.30 ms           | 4.46 ms   | 100%       | 15      | 100.0%    |
| <b>cache_hit_small_logs</b> · wrk · 64 conn · cache_hit_small with the access log on in both: zion’s default, nginx’s combined format |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 29.1k                                                                                     | 28.5k–30.0k | 34.3                  | 2.17 ms           | 2.53 ms           | 3.74 ms   | 100%       | 25      | 100.0%    |
| zion                                                                                                                                  | 19.1k                                                                                     | 18.4k–19.1k | 52.4                  | 3.37 ms           | 4.98 ms           | 5.72 ms   | 100%       | 15      | 100.0%    |
| <b>download_10m</b> · wrk · 8 conn · a 10 MiB body streamed through the proxy, 8 connections                                          |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 69.9                                                                                      | 69.5–70.8   | 14.3k                 | 113.73 ms         | 126.69 ms         | 142.85 ms | 100%       | 23      | –         |
| origin                                                                                                                                | 394.5  | 387.8–404.9 | 763.5                 | 14.68 ms          | 24.79 ms          | 39.81 ms  | 30%        | 13      | –         |
| zion                                                                                                                                  | 129.0                                                                                     | 127.3–129.4 | 7.8k                  | 64.12 ms          | 86.60 ms          | 112.52 ms | 100%       | 26      | –         |
| <b>latency_15k</b> · oha · 50 conn · a fixed 15000 req/s on the cached document, latency corrected for coordinated omission           |                                                                                           |             |                       |                   |                   |           |            |         |           |
| nginx                                                                                                                                 | 15.0k                                                                                     | 15.0k–15.0k | 32.3                  | 477 $\mu\text{s}$ | 7.94 ms           | 35.20 ms  | 48%        | 24      | 100.0%    |
| origin                                                                                                                                | 15.0k                                                                                     | 15.0k–15.0k | 21.1                  | 304 $\mu\text{s}$ | 730 $\mu\text{s}$ | 2.48 ms   | 31%        | 13      | –         |
| zion                                                                                                                                  | 15.0k                                                                                     | 15.0k–15.0k | 30.6                  | 488 $\mu\text{s}$ | 3.41 ms           | 30.34 ms  | 46%        | 14      | 100.0%    |

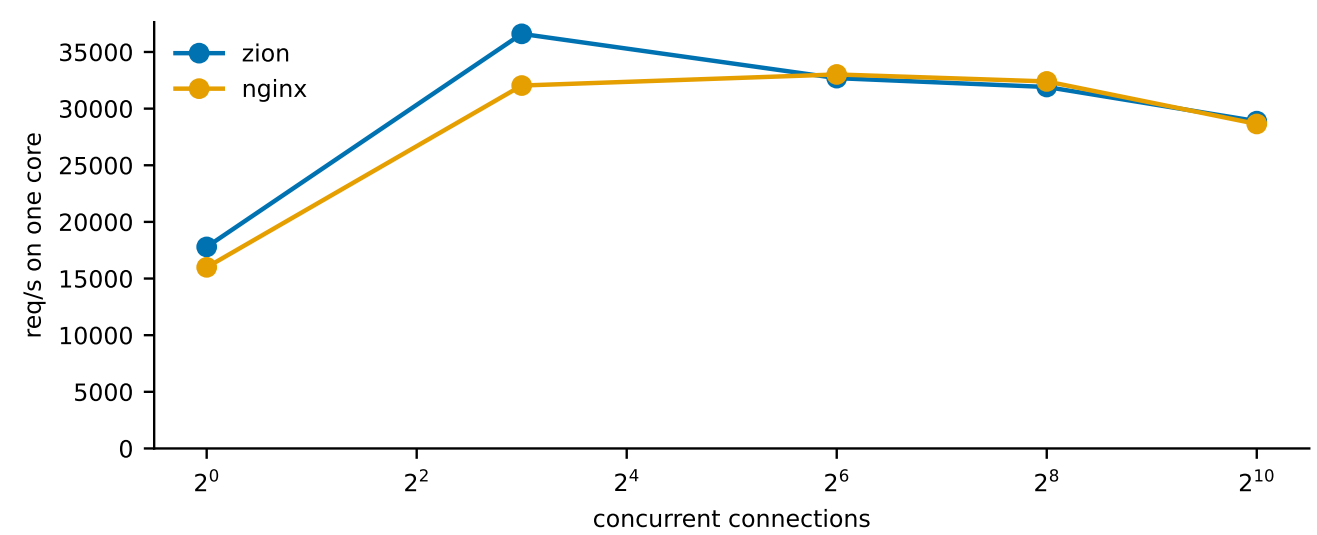
|                                                                                                                           |                                                                                           |             |       |          |           |           |      |    |        |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------|-------|----------|-----------|-----------|------|----|--------|
| <b>latency_5k</b> · oha · 50 conn · a fixed 5000 req/s on the cached document, latency corrected for coordinated omission |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 5.0k                                                                                      | 5.0k–5.0k   | 44.4  | 271 µs   | 872 µs    | 15.01 ms  | 22%  | 24 | 100.0% |
| origin                                                                                                                    | 5.0k                                                                                      | 5.0k–5.0k   | 30.8  | 214 µs   | 493 µs    | 1.98 ms   | 15%  | 13 | –      |
| zion                                                                                                                      | 5.0k                                                                                      | 5.0k–5.0k   | 43.2  | 264 µs   | 755 µs    | 21.57 ms  | 22%  | 14 | 100.0% |
| <b>proxy_nocache</b> · wrk · 64 conn · no cache: every request goes to the origin (origin: no-store)                      |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 24.6k                                                                                     | 24.2k–24.7k | 40.5  | 2.56 ms  | 3.89 ms   | 7.41 ms   | 99%  | 25 | –      |
| origin                                                                                                                    | 54.9k                                                                                     | 54.3k–56.1k | 18.2  | 1.16 ms  | 1.45 ms   | 3.01 ms   | 100% | 13 | –      |
| zion                                                                                                                      | 17.6k                                                                                     | 17.1k–17.9k | 56.4  | 3.64 ms  | 5.03 ms   | 9.71 ms   | 99%  | 17 | –      |
| <b>site_mix</b> · wrk · 64 conn · the whole example site, Zipf-popular paths, hot cache                                   |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 11.2k                                                                                     | 11.0k–11.4k | 89.1  | 5.53 ms  | 9.68 ms   | 11.46 ms  | 100% | 25 | 100.0% |
| origin                                                                                                                    | 43.6k                                                                                     | 43.3k–44.0k | 22.8  | 1.35 ms  | 3.01 ms   | 36.78 ms  | 99%  | 13 | –      |
| zion                                                                                                                      | 15.9k                                                                                     | 15.8k–16.2k | 62.8  | 3.98 ms  | 6.30 ms   | 7.93 ms   | 100% | 39 | 100.0% |
| <b>site_mix_logs</b> · wrk · 64 conn · site_mix with the access log on in both                                            |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 10.6k                                                                                     | 9.8k–10.7k  | 94.7  | 5.88 ms  | 10.18 ms  | 12.19 ms  | 100% | 26 | 100.0% |
| zion                                                                                                                      | 11.3k                                                                                     | 10.5k–11.3k | 88.3  | 5.65 ms  | 8.34 ms   | 9.64 ms   | 100% | 40 | 100.0% |
| <b>static_files</b> · wrk · 64 conn · zion `static` mode and nginx `alias` serving the files from disk                    |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 20.7k                                                                                     | 20.1k–20.9k | 48.4  | 3.07 ms  | 3.52 ms   | 5.03 ms   | 100% | 25 | –      |
| origin                                                                                                                    | 57.6k                                                                                     | 57.3k–57.9k | 17.3  | 1.08 ms  | 1.62 ms   | 3.29 ms   | 100% | 13 | –      |
| zion                                                                                                                      | 8.8k                                                                                      | 8.7k–8.8k   | 113.9 | 7.15 ms  | 17.00 ms  | 19.85 ms  | 100% | 27 | –      |
| <b>sweep_c1</b> · wrk · 1 conn · the cached document at 1 connections                                                     |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 16.0k  | 15.6k–16.8k | 37.1  | 51 µs    | 778 µs    | 2.10 ms   | 60%  | 23 | 100.0% |
| zion                                                                                                                      | 17.8k  | 14.8k–17.9k | 32.9  | 47 µs    | 986 µs    | 2.11 ms   | 58%  | 12 | 100.0% |
| <b>sweep_c1024</b> · wrk · 1024 conn · the cached document at 1024 connections                                            |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 28.6k                                                                                     | 27.8k–29.4k | 34.7  | 33.88 ms | 53.03 ms  | 230.26 ms | 99%  | 67 | 100.0% |
| zion                                                                                                                      | 28.9k                                                                                     | 28.6k–29.2k | 34.2  | 9.52 ms  | 177.03 ms | 236.38 ms | 99%  | 57 | 100.0% |
| <b>sweep_c256</b> · wrk · 256 conn · the cached document at 256 connections                                               |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 32.4k                                                                                     | 30.6k–32.5k | 30.7  | 7.69 ms  | 9.75 ms   | 26.20 ms  | 99%  | 34 | 100.0% |
| zion                                                                                                                      | 31.9k                                                                                     | 31.2k–33.9k | 31.1  | 7.94 ms  | 9.77 ms   | 42.73 ms  | 99%  | 23 | 100.0% |
| <b>sweep_c8</b> · wrk · 8 conn · the cached document at 8 connections                                                     |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 32.0k                                                                                     | 31.5k–32.7k | 31.0  | 239 µs   | 418 µs    | 2.10 ms   | 99%  | 23 | 100.0% |
| zion                                                                                                                      | 36.6k                                                                                     | 36.2k–36.9k | 27.2  | 230 µs   | 403 µs    | 2.10 ms   | 100% | 12 | 100.0% |
| <b>tls_handshake</b> · wrk · 64 conn · Connection: close, so every request is a new TLS handshake                         |                                                                                           |             |       |          |           |           |      |    |        |
| nginx                                                                                                                     | 2.4k                                                                                      | 2.4k–2.4k   | 410.3 | 859 µs   | 1.75 ms   | 12.31 ms  | 100% | 25 | 100.0% |
| zion                                                                                                                      | 3.2k                                                                                      | 3.2k–3.2k   | 308.8 | 9.24 ms  | 11.20 ms  | 20.51 ms  | 99%  | 14 | 100.0% |

Latencies for wrk are measured under saturation (64 concurrent connections), so they show queueing, not service time; the fixed-rate scenarios below show what a user would feel. h2load reports no percentiles.

## Latency at a fixed request rate



## Concurrency sweep



## HTTP conformance

Every check is a fixed request with a fixed expectation, run against zion and, as a reference, against nginx proxying the same origins. The probe origin sends the malformed and edge-case messages a real origin will not, and records what the proxy forwarded.

Failed **MUST** or **SECURITY** checks: **zion 4**, nginx 4 (the reference, same origin, same checks). A failed **SHOULD** is a deviation to read, not a defect; an **INFO** row records behaviour and cannot fail.

| Group           | Checks | zion pass | zion fail | nginx pass | nginx fail |
|-----------------|--------|-----------|-----------|------------|------------|
| Framing         | 24     | 18        | 3         | 20         | 1          |
| Representations | 12     | 12        | 0         | 11         | 1          |
| TLS             | 4      | 2         | 0         | 2          | 0          |
| Hardening       | 4      | 4         | 0         | 4          | 0          |
| Proxy           | 11     | 7         | 1         | 4          | 4          |

| Check                                                                                              | Level | RFC             | zion | nginx | What zion did           |
|----------------------------------------------------------------------------------------------------|-------|-----------------|------|-------|-------------------------|
| Host with a space in it: 400                                                                       | MUST  | RFC 9112 §3.2   | fail | pass  | status 200              |
| Bare CR in a field value: rejected or replaced                                                     | MUST  | RFC 9112 §2.2   | pass | pass  | rejected with 400       |
| Malformed chunk size: not accepted                                                                 | MUST  | RFC 9112 §7.1   | pass | pass  | status 502              |
| A chunked request body arrives at the origin byte for byte                                         | MUST  | RFC 9112 §7.1   | pass | pass  | status 200              |
| Malformed Content-Length (+5, '5 5', 0x5, -1, 5abc): rejected, not forwarded                       | MUST  | RFC 9110 §8.6   | pass | pass  | all five rejected       |
| Content-Length and Transfer-Encoding together: not forwarded as both, and the connection is closed | MUST  | RFC 9112 §6.1   | pass | pass  | answered and closed     |
| Connection: close is honoured                                                                      | MUST  | RFC 9112 §9.6   | pass | pass  | status 200, closed True |
| Two different Content-Length values: rejected, not forwarded                                       | MUST  | RFC 9110 §8.6   | pass | pass  | status 400, forwarded 0 |
| Lower-case method name is not a method: not served                                                 | MUST  | RFC 9110 §9.1   | pass | pass  | status 405              |
| HTTP/1.1 request without Host: 400                                                                 | MUST  | RFC 9112 §3.2   | fail | pass  | status 200              |
| NUL in a field value: rejected or replaced                                                         | MUST  | RFC 9110 §5.5   | pass | pass  | rejected with 400       |
| Obsolete line folding: rejected or replaced by a space                                             | MUST  | RFC 9112 §5.2   | pass | pass  | rejected with 400       |
| Two pipelined requests are answered in order                                                       | MUST  | RFC 9112 §9.3.2 | pass | pass  | 2 responses             |
| Transfer-Encoding: chunked, gzip (chunked not last): 400                                           | MUST  | RFC 9112 §6.3   | pass | fail  | status 400, forwarded 0 |
| Unknown transfer coding (xchunked): 400 or 501, not forwarded                                      | MUST  | RFC 9112 §6.1   | pass | pass  | status 400, forwarded 0 |
| Two Host header lines: 400                                                                         | MUST  | RFC 9112 §3.2   | fail | pass  | status 200              |
| Whitespace before the colon in a field name: 400                                                   | MUST  | RFC 9112 §5.1   | pass | pass  | status 400, forwarded 0 |

|                                                                                                |        |                   |      |      |                                                       |
|------------------------------------------------------------------------------------------------|--------|-------------------|------|------|-------------------------------------------------------|
| Expect: 100-continue: the body is delivered and answered                                       | SHOULD | RFC 9110 §10.1.1  | pass | pass | status 200, interim [100]                             |
| 200 KB of request headers: 431 (or another 4xx), not a 5xx or a hang                           | SHOULD | RFC 9110 §15.5.32 | pass | pass | status 431                                            |
| A 70,000-octet request target gets 414 (or another 4xx), not a crash                           | SHOULD | RFC 9110 §15.5.15 | pass | pass | status 431                                            |
| A request line of 8000 octets is accepted                                                      | SHOULD | RFC 9112 §3       | pass | pass | status 200                                            |
| Absolute-form target with a different Host header                                              | INFO   | RFC 9112 §3.2.2   | info | info | status 200; origin saw Host '127.0.0.1:14491'         |
| HTTP/1.0 request without Host                                                                  | INFO   | RFC 9112 §3.2     | info | info | status 200                                            |
| OPTIONS *                                                                                      | INFO   | RFC 9110 §9.3.7   | info | info | status 404                                            |
| Hop-by-hop request headers (Connection-listed, Keep-Alive, Proxy-Connection) are not forwarded | MUST   | RFC 9110 §7.6.1   | fail | fail | leaked: x-remove-me                                   |
| Hop-by-hop response headers (Connection-listed, Keep-Alive) are not passed on                  | MUST   | RFC 9110 §7.6.1   | pass | fail | status 200, none leaked                               |
| A 100 Continue before the final response does not confuse the client                           | MUST   | RFC 9110 §15.2    | pass | pass | status 200, body b'ok'                                |
| An origin response with both Content-Length and Transfer-Encoding is not relayed as both       | MUST   | RFC 9112 §6.1     | pass | pass | status 200, both headers False, body b'hello'         |
| A response cut short by the origin is not delivered as complete                                | MUST   | RFC 9112 §8       | pass | pass | status 200, complete False, body 10 bytes             |
| A response header line without a colon is not relayed                                          | SHOULD | RFC 9110 §5.5     | pass | fail | status 502, relayed False                             |
| An unreachable origin gives 502/503/504 promptly                                               | SHOULD | RFC 9110 §15.6.3  | pass | pass | status 503 after 0.0s                                 |
| A Via header is added to the forwarded request                                                 | SHOULD | RFC 9110 §7.6.3   | pass | fail | Via: 1.1 zion-5b11d3a5                                |
| OPTIONS with Max-Forwards: 0                                                                   | INFO   | RFC 9110 §7.6.2   | info | info | status 200; forwarded 1                               |
| A 60 KB response header                                                                        | INFO   | RFC 9110 §5.4     | info | info | status 200                                            |
| X-Forwarded-For from the client: what the origin sees                                          | INFO   | (not an RFC)      | info | info | origin saw X-Forwarded-For: '127.0.0.1'               |
| A response served from cache carries Age                                                       | MUST   | RFC 9111 §4       | pass | fail | Age: 0                                                |
| If-None-Match with the current ETag: 304, no body, validator kept                              | MUST   | RFC 9110 §13.1.2  | pass | pass | status 304, ETag "6ac80882-c23"                       |
| HEAD: same status and Content-Length as GET, no body                                           | MUST   | RFC 9110 §9.3.2   | pass | pass | HEAD 200 CL 3107 vs GET 200 CL 3107                   |
| If-Modified-Since with Last-Modified: 304                                                      | MUST   | RFC 9110 §13.1.3  | pass | pass | status 304                                            |
| If-None-Match with a different ETag: full 200                                                  | MUST   | RFC 9110 §13.1.2  | pass | pass | status 200                                            |
| Range: bytes=0-99: a correct 206, or the whole 200 (both allowed)                              | MUST   | RFC 9110 §14      | pass | pass | status 206, Content-Range bytes 0-99/29266, 100 bytes |
| Static files answer If-None-Match with 304                                                     | MUST   | RFC 9110 §13      | pass | pass | status 304                                            |
| POST to a cached URL is not answered from the cache                                            | MUST   | RFC 9111 §4.4     | pass | pass | status 405 (origin says 405), Age None                |
| Static files advertise Accept-Ranges: bytes                                                    | SHOULD | RFC 9110 §14.3    | pass | pass | status 200, Accept-Ranges bytes                       |

|                                                                            |          |                   |      |      |                                                     |
|----------------------------------------------------------------------------|----------|-------------------|------|------|-----------------------------------------------------|
| The origin's Content-Type is preserved                                     | SHOULD   | RFC 9110 §8.3     | pass | pass | Content-Type: application/json                      |
| An unsatisfiable range: 416 with Content-Range: bytes */N                  | SHOULD   | RFC 9110 §15.5.17 | pass | pass | status 416, Content-Range bytes */29266             |
| Static files honour Range                                                  | SHOULD   | RFC 9110 §14      | pass | pass | status 206, 10 bytes                                |
| CONNECT does not open a tunnel                                             | SECURITY | RFC 9110 §9.3.6   | pass | pass | status 405                                          |
| PUT and DELETE on the static directory do not succeed                      | SECURITY | RFC 9110 §9.3     | pass | pass | all refused                                         |
| TRACE is not forwarded to the origin                                       | SECURITY | RFC 9110 §9.3.8   | pass | pass | status 405, forwarded 0                             |
| Path traversal out of the static directory never reaches a file outside it | SECURITY | RFC 3986 §6       | pass | pass | 12 encodings, none reached the file                 |
| A session is resumed with a ticket                                         | SHOULD   | RFC 8446 §4.6.1   | pass | pass | session_reused=True                                 |
| TLS 1.3 is negotiated                                                      | SHOULD   | RFC 8446          | pass | pass | TLSv1.3                                             |
| ALPN selection when the client offers h2 and http/1.1                      | INFO     | RFC 7301          | info | info | selected h2                                         |
| Whether TLS 1.2 is still accepted (a configured floor, not a requirement)  | INFO     | RFC 5246          | info | info | refused (tls1 alert protocol version (_ssl.c:1000)) |

## HTTP/2 — h2spec (RFC 9113, RFC 7541)

|       | tests | passed | failed | skipped |
|-------|-------|--------|--------|---------|
| zion  | 146   | 135    | 7      | 4       |
| nginx | 146   | 119    | 27     | 0       |

zion failures:

- [generic/2] Sends a WINDOW\_UPDATE frame on half-closed (remote) stream — DATA frame
- [generic/2] Sends a PRIORITY frame on half-closed (remote) stream — DATA frame
- [http2/3.5] Sends invalid connection preface — GOAWAY Frame (Error Code: PROTOCOL\_ERROR)
- [http2/5.1] closed: Sends a DATA frame after sending RST\_STREAM frame — GOAWAY Frame (Error Code: STREAM\_CLOSED)
- [http2/5.1.2] Sends HEADERS frames that causes their advertised concurrent stream limit to be exceeded — GOAWAY Frame (Error Code: PROTOCOL\_ERROR)
- [http2/8.1] Sends a second HEADERS frame without the END\_STREAM flag — GOAWAY Frame (Error Code: PROTOCOL\_ERROR)
- [http2/8.1.2.1] Sends a HEADERS frame that contains a pseudo-header field as trailers — GOAWAY Frame (Error Code: PROTOCOL\_ERROR)

## HTTP caching behaviour — cache-tests (RFC 9111)

The suite describes itself as “not a conformance test suite ... a tool to assess how a cache behaves”: it also probes optional behaviour. Counts are shown for both proxies on the same origin; read the differences, not the totals.

|       | tests | passed | failed |
|-------|-------|--------|--------|
| zion  | 370   | 100    | 51     |
| nginx | 370   | 195    | 36     |

## Byte-exact serving of the example site

230 files fetched twice (cold and from cache) through zion, compared by SHA-256 with the manifest: **0 mismatches**.

## Against the previous release

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No previous report was given, so there is no version-to-version comparison. Pass `--prev` to add one.

## Provenance and how to repeat this

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Tools (pinned in `benchmarks/report/provision.sh`):

```
cache_tests d644cf4bf487763646aac19d2c8b846daa0f604d
h2load      h2load nghhttp2/1.59.0
h2spec      Version: 2.6.0 (70ac2294010887f48b18e2d64f5cccd48421fad1)
kernel      6.17.4-2-pve
libc        ldd (Ubuntu GLIBC 2.39-0ubuntu8.9) 2.39
matplotlib  3.11.2
nginx       nginx version: nginx/1.24.0 (Ubuntu)
node        v20.20.2
oha         oha 1.16.0
python      Python 3.12.3
weasyprint  70.0
wrk         wrk debian/4.1.0-4build2 [epoll] Copyright (C) 2012 Will Glozer
```

Canary (SHA-256 MB/s on the server core, start → end): 474, 474, 472, 471, 463, 485.

```
benchmarks/report/provision.sh      # once, on the benchmark host
benchmarks/report/report.sh v0.11.0 # build the tag, measure, test, render this PDF
```